



Nakajima Ki-27

The **Nakajima Ki-27** (九七式戦闘機, *Kyūnana-shiki sentōki*; Type 97 Fighter) was the main fighter aircraft used by the Imperial Japanese Army Air Service up until 1940. Its Allied nickname was "**Nate**", although it was called "**Abdul**" in the "China Burma India" (CBI) theater by many post-war sources;^[1] Allied Intelligence had reserved that name for the nonexistent Mitsubishi Navy Type 97 fighter, expected to be the successor to the carrier-borne Type 96 (Mitsubishi A5M) with retractable landing gear and an enclosed cockpit.^[2]

Design and development

In 1935, the Imperial Japanese Army held a competition between Nakajima, Mitsubishi, and Kawasaki to design a low-wing monoplane to replace the Kawasaki Ki-10 (Army Type 95 Fighter) biplane. The new fighter was to have also a better performance than the experimental Mitsubishi Ki-18.^[3]

The results were the Nakajima Ki-27, the Kawasaki Ki-28, and the Mitsubishi Ki-33 (a modification of the Mitsubishi A5M carrier-based fighter).^{[4][5]} The Nakajima design was based on its earlier Ki-11 monoplane fighter which lost to the Ki-10 in the Type 95 Fighter competition. When the follow-up Nakajima Ki-12 proposal with a liquid-cooled engine and retractable landing gear was deemed too complex by the Japanese officials, the Ki-27 was designed by Koyama Yasushi to have an air-cooled radial engine and fixed landing gear. The aircraft had the Nakajima trademark wing with a straight leading edge and tapered trailing edge which would reappear again on the Ki-43, Ki-44, and Ki-84.

The Ki-27 made its first flight on 15 October 1936.^{[6][7]} Although it had a slower top speed and worse climb performance than its competitors,^[8] the Army chose the Nakajima design for its outstanding turning ability granted by

Ki-27



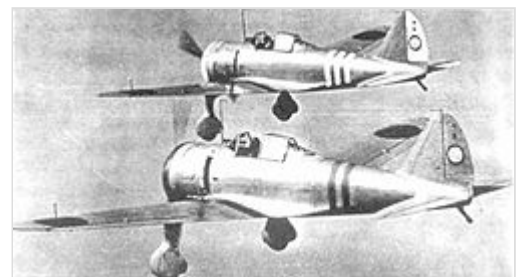
Nakajima Ki-27

General information

Type	<u>Fighter aircraft</u>
National origin	<u>Japan</u>
Manufacturer	<u>Nakajima Aircraft Company</u>
Primary users	<u>Imperial Japanese Army Air Service</u> <u>Manchukuo Air Force</u> <u>Royal Thai Air Force</u> <u>Reformed Government of the Republic of China</u>
Number built	3,368

History

Introduction date	1937
First flight	15 October 1936
Retired	1945 (Japan)



Nakajima Ki-27 of the Akeno Army Flying School, ca. winter 1941/42 (see Bueschel 1970)

its remarkably low wing loading. The Army ordered 10 pre-production samples (Ki-27a) for further testing, which featured an enclosed cockpit with sliding canopy and larger wings.

The type was officially accepted into service in 1937 as the **Army Type 97 Fighter**. In addition to Nakajima, the Ki-27 was also manufactured by Tachikawa Aircraft Company Ltd and Manshukoku Hikoki Seizo KK, with a total of 3,368 built before production ended in 1942.

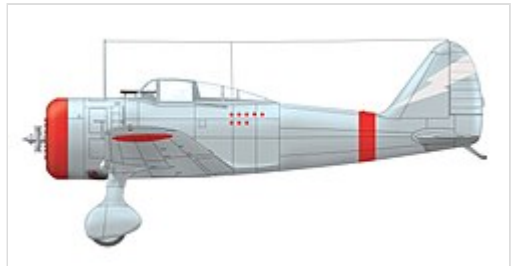
Operational history

The Ki-27 was the Imperial Japanese Army Air Force's main fighter until the start of World War II. When placed into combat service over northern China in March 1938, the Ki-27 enjoyed air superiority until the introduction of the faster Soviet-built Polikarpov I-16 fighters by the Chinese.

In the 1939 Battle of Khalkhin Gol against the USSR in Mongolia, the Ki-27 faced both Polikarpov I-15 biplane and Polikarpov I-16 monoplane fighters. In the initial phase of the conflict, its performance was a match for the early model I-16s, and was considerably superior to the I-15 biplane. With better trained Ki-27 pilots, the IJAAF gained aerial superiority. The Ki-27 was armed with two 7.7 mm (.303 in) Type 89 machine guns and as with most aircraft of the period, lacked armor protection for the pilot and self-sealing or fire suppression in the fuel tanks.

Later, the Soviet Air Force received improved I-16s. The faster, more heavily armed (with twin wing-mounted 20mm ShVAK cannon) and armored I-16 now nullified the Ki-27's advantages and it could now escape from the Ki-27 in a dive.

The VVS introduced new tactics consisting of flying in large tightly knit formations, attacking with altitude and/or speed advantage and hit-and-run (high-energy) tactics much as Claire Chennault would later formulate for the 1941-era Flying Tigers (likewise to fly against Japanese forces).



Nakajima Ki-27b of Kenji Shimada, commander of the 1st Chutai of the 11th Sentai, Battle of Khalkhin Gol June 1939



A Ki-27 as used in the Battle of Khalkhin Gol.



Ki-27s at Nomonhan, 1939

Japanese losses mounted but despite this they claimed 1,340 aircraft (six times the admitted Soviet losses and three times as many as Soviet aircraft admitted to being in the theatre).^[9] Japanese losses numbered 120 (including Ki-10s) while the Russians claimed 215 vs. a peak Japanese strength of 200 fighters.^[9] (Overclaiming remained commonplace through World War Two, despite gun cameras and expert intelligence assessments.) Top scoring pilot of the incident and top scoring IJAAF pilot on the Ki-27 and overall World War II IJAAF ace was Warrant Officer Hiromichi Shinohara, who claimed 58 Soviet planes (including an IJAAF record of 11 in one day) whilst flying Ki-27s, only to be shot down himself by a

number of I-16s on 27 August 1939.^[10]

The preference of Japanese fighter pilots for the Ki-27's high rate of turn caused the Army to focus excessively on manoeuvrability, a decision which later handicapped the development of faster and more heavily armed fighters. The Ki-27 served until the beginning of World War II in the Pacific, escorting bombers attacking Malaya, Singapore, Netherlands East Indies, Burma and the Philippines (where it initially fared poorly against the Brewster F2A Buffalo).^[11]



Ki-27 replica at Tokorozawa Aviation Museum

The type also saw extensive action against the American Volunteer Group in the early months of the war. Soon outclassed by the American Curtiss P-40 Warhawks, the Ki-27 was replaced in front line service by the Nakajima Ki-43, while surviving examples continued to serve as trainers.

The Ki-27 was also exported for use with Manchukuo and Thai armed forces, seeing combat with both. In Thai service, Ki-27s reportedly damaged two North American P-51 Mustangs and shot down one Lockheed P-38 Lightning and one North American P-51 Mustang.^[12]

In the final months of the war, desperate lack of aircraft forced the Japanese to utilize all available machines, and the Ki-27 and 79 were no exception. Some were equipped with up to 500 kg (1,100 lb) of explosives for kamikaze attacks, but some were redeployed as fighters, suffering terrible losses. On 16 February 1945, the 39th Educational Flight Regiment scrambled 16 Ki-79 trainers from Yokoshiba Airfield to oppose a massive air raid from U.S. Task Force 58 carrier group, losing six aircraft with more damaged and five pilots killed - in return damaging at least one F6F Hellcat and possibly downing a second.^[13]

Variants

Data from^[14]

Nakajima Army Type 97 Fighter

Long Army designation for the Ki-27

Nakajima Type PE

Private-venture experimental aircraft with Nakajima Ha1a engine.

Nakajima Ki-27

Prototype version with armament in response to IJAAF specs, two aircraft built.

Nakajima Ki-27-Kai Prototype

Pre-production units with armament and heavier Nakajima Ha1b engine, 10 aircraft built.

Ki-27a

First production version. Approximately 565 aircraft built.

Ki-27a-Kai

Trainer version converted from existing production. Approximately 150 aircraft converted.

Ki-27b (Army Type 97b Fighter)



Japanese army personnel and their Ki-27s

Improved canopy, oil cooler and provision for 4 × 25 kg (55 lb) bombs or fuel tanks under the wings. A total of 1,492 built, including 50 by Tachikawa Aircraft Company Ltd.

Ki-27b-Kai

Trainer version converted from existing production. Approximately 225 aircraft converted.

Nakajima Ki-27-Kai

Experimental lightened version developed as an interim solution when Ki-43 development was delayed, top speed 475 km/h (295 mph); two aircraft built^[15].

Mansyū Ki-79

Trainer version, built by Manshūkoku Hikōki Seizo KK with a 510 hp Hitachi Ha.13a-I or Ha.13a-III engine. A total of 1,329 aircraft built in four sub-versions (The single seat Ki-79a (Ha.13a-I) and Ki-79c (Ha.13a-III) and the two-seat Ki-79b (Ha.13a-I) and Ki-79d (Ha.13a-III)^{[16][17]}).

Mansyū Army Type 2 Advanced Trainer

Long Army designation for the Mansyū Ki-79

B.Kh.12

(Thai: อ.อ.๑๒) Royal Thai Air Force designation for the Ki-27b.^[18]

Operators

World War II

● Japan

- Imperial Japanese Army Air Service^{[19][20][21]}
 - No. 2 Dokuritsu Hikō Daitai IJAAF
 - No. 9 Dokuritsu Hikō Chutai IJAAF
 - No. 10 Dokuritsu Hikō Chutai IJAAF
 - No. 84 Dokuritsu Hikō Chutai IJAAF
 - No. 102 Dokuritsu Hikō Chutai IJAAF
 - No. 1 Hikō Sentai IJAAF
 - No. 2 Hikō Sentai IJAAF
 - No. 4 Hikō Sentai IJAAF
 - No. 5 Hikō Sentai IJAAF
 - No. 9 Hikō Sentai IJAAF
 - No. 11 Hikō Sentai IJAAF
 - No. 13 Hikō Sentai IJAAF
 - No. 18 Hikō Sentai IJAAF
 - No. 21 Hikō Sentai IJAAF
 - No. 24 Hikō Sentai IJAAF
 - No. 26 Hikō Sentai IJAAF
 - No. 29 Hikō Sentai IJAAF
 - No. 30 Hikō Sentai IJAAF
 - No. 33 Hikō Sentai IJAAF
 - No. 48 Hikō Sentai IJAAF
 - No. 50 Hikō Sentai IJAAF
 - No. 54 Hikō Sentai IJAAF

- [No. 59 Hikō Sentai IJAAF](#)
- [No. 63 Hikō Sentai IJAAF](#)
- [No. 64 Hikō Sentai IJAAF](#)
- [No. 68 Hikō Sentai IJAAF](#)
- [No. 70 Hikō Sentai IJAAF](#)
- [No. 77 Hikō Sentai IJAAF](#)
- [No. 78 Hikō Sentai IJAAF](#)
- [No. 85 Hikō Sentai IJAAF](#)
- [No. 87 Hikō Sentai IJAAF](#)
- [No. 101 Hikō Sentai IJAAF](#)
- [No. 144 Hikō Sentai IJAAF](#)
- [No. 204 Hikō Sentai IJAAF](#)
- [No. 206 Hikō Sentai IJAAF](#)
- [No. 244 Hikō Sentai IJAAF](#)
- [No. 246 Hikō Sentai IJAAF](#)
- [No. 248 Hikō Sentai IJAAF](#)
- [Rikugun Koukuu Shikan Gakkō](#)
- [Tokorozawa Rikugun Koku Seibi Gakkō](#)
- [Akeno Rikugun Hikō Gakkō](#)
- [Kumagaya Rikugun Hikō Gakkō](#)
- [Tachiarai Rikugun Hikō Gakkō](#)

Manchukuo

- [Manchukuo Air Force](#)

Reformed Government of the Republic of China

- Scheduled aircraft never delivered due to distrust of Chinese forces^[22]

Thailand

- [Royal Thai Air Force](#)^[23]
 - Foong Bin Khap Lai 15 (15 Fighter Squadron)^[12]
 - Foong Bin Khap Lai 16 (16 Fighter Squadron)^[24]

Post-War

China

- [People's Liberation Army Air Force](#)

China

- [Republic of China Air Force](#)

Indonesia



Ki-27 in ROCAF markings



A Nakajima Ki-27 in [Tachiarai Peace Memorial Museum](#)

- In 1945, Indonesian Air Force –then Indonesian People's Security Force (IPSF) (Indonesian pro-independence guerrillas)– captured a small number of aircraft at numerous Japanese air bases, including Bugis Air Base in Malang (repatriated 18 September 1945). Most aircraft were destroyed in military conflicts between the Netherlands and the newly proclaimed-Republic of Indonesia during the Indonesian National Revolution of 1945–1949.^[25]



Mansyū Ki-79 in Museum Dirgantara Mandala

Surviving aircraft

Two aircraft survive today:

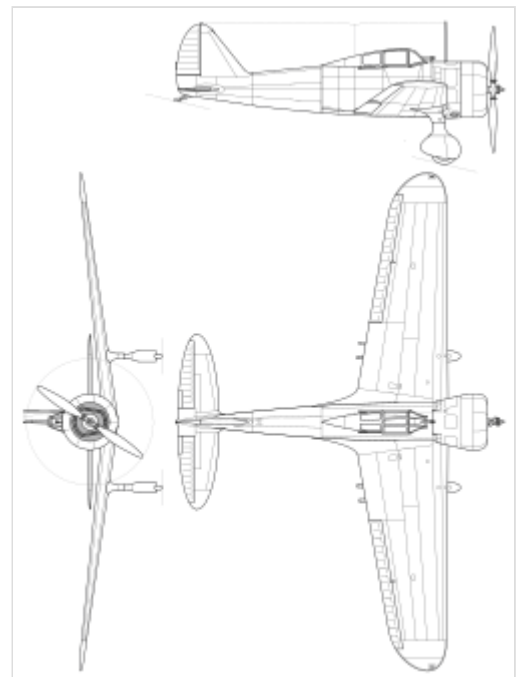
- One Ki-27 is preserved at the Tachiarai Peace Memorial Museum.^{[26][27]}
- One Mansyu Ki-79 is preserved at the Satria Mandala Armed Forces Museum, Jakarta, Indonesia.^[28]

Specifications (Ki-27b)

Data from Nakajima Ki-27 Nate,^[20] Japanese Aircraft of the Pacific War^[29]

General characteristics

- **Crew:** One
- **Length:** 7.53 m (24 ft 8 in)
- **Wingspan:** 11.31 m (37 ft 1 in)
- **Height:** 3.25 m (10 ft 8 in)
- **Wing area:** 18.56 m² (199.8 sq ft)
- **Airfoil:** **root:** NN-2 mod (16%); **tip:** NN-2 mod (8%)^[30]
- **Empty weight:** 1,110 kg (2,447 lb)
- **Gross weight:** 1,790 kg (3,946 lb)
- **Powerplant:** 1 × Nakajima Ha-1 Kotobuki Otsu (Ha-1b) 9-cylinder air-cooled radial piston engine, 530 kW (710 hp) for take-off, 582 kW (780 hp) at 2,900 m (9,500 ft)
- **Propellers:** 2-bladed variable-pitch propeller



3-view drawing of Nakajima Ki-27

Performance

- **Maximum speed:** 470 km/h (290 mph, 250 kn) at 3,500 m (11,500 ft)
- **Cruise speed:** 350 km/h (220 mph, 190 kn) at 3,500 m (11,500 ft)
- **Range:** 630 km (390 mi, 340 nmi)
- **Ferry range:** 1,710 km (1,060 mi, 920 nmi)
- **Time to altitude:** 5,000 m (16,000 ft) in 5 minutes 22 seconds (3,056 fpm)
- **Wing loading:** 96.5 kg/m² (19.8 lb/sq ft) Maximum weight
- **Power/mass:** 3.43 kg/kW (5.64 lb/hp) maximum

Armament

■ Guns:

- 2 × 7.7 mm (0.303 in) Type 89 machine guns, 500 rounds/gun
- 1 × 12.7 mm (0.5 in) Ho-103 machine gun and 1 × 7.7 mm (0.303 in) machine gun on later models

■ Bombs:

- 4 × 25 kg (55 lb)
- 2 × 130 L (34 US gal; 29 imp gal) drop-tanks

See also

Related development

- Nakajima Ki-11
- Nakajima Ki-12
- Nakajima Ki-43

Aircraft of comparable role, configuration, and era

- Arado Ar 80
- Curtiss P-36 Hawk
- Dewoitine D.500
- Fokker D.XXI
- Gloster F.5/34
- Polikarpov I-16

Related lists

- List of aircraft of Japan during World War II
- List of aircraft of World War II
- List of military aircraft of Japan

References

Notes

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4. Januszewski 2003, pp. 10–11.
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6. Francillon 1979, p. 198.
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9. Francillon 1979, p. 200.
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12. Wieliczko & Szeremeta 2004, p. 81.
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21. Thorpe 1968, pp. 1881–185.
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